BridgeWell

EMPLOYEE SUPPORT SOLUTIONS

Diana Freeman, Sruthi Papanasa, Mitra Rezvany, Natalie Tehranchi

DSC202 Final Project

75%

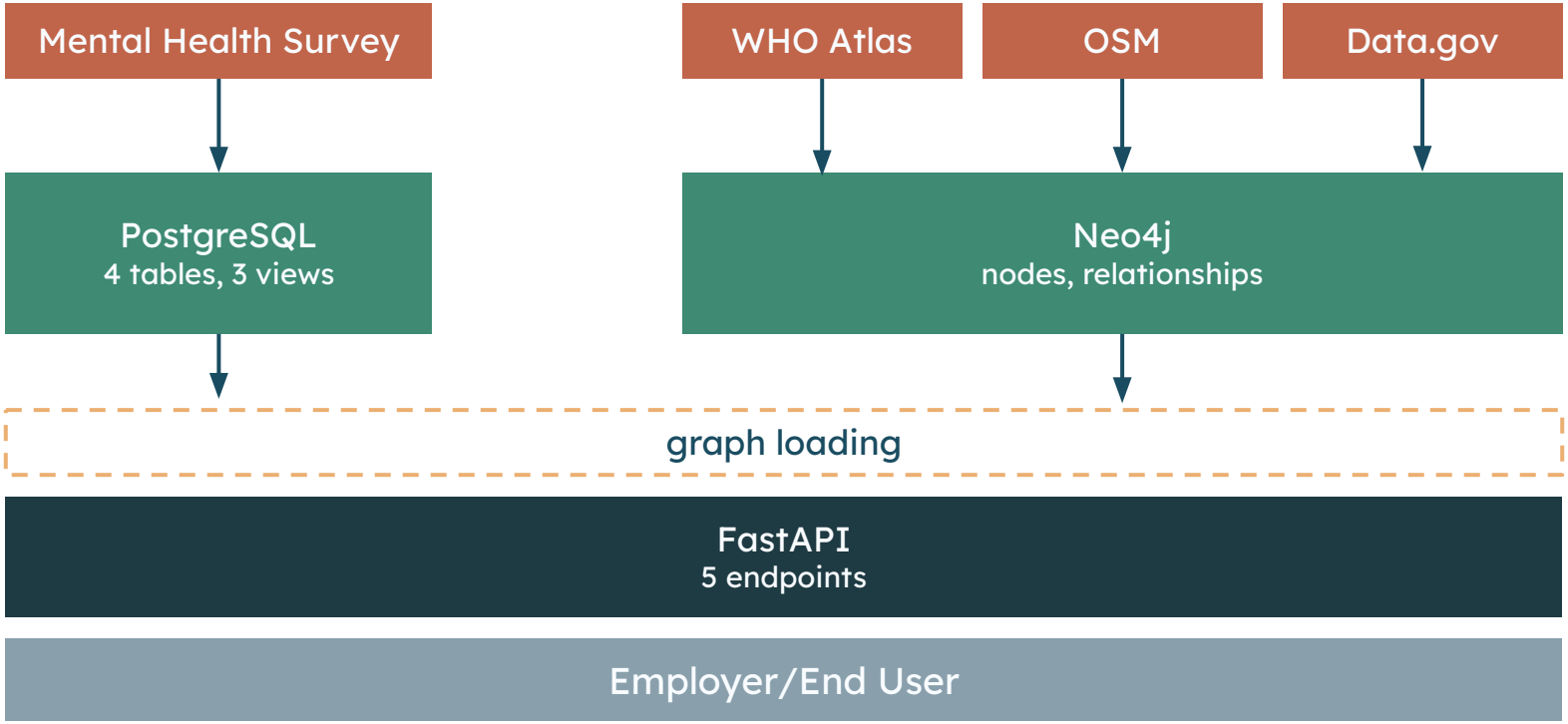
Of workers across every country lack mental health support at work.[1]

Companies want to help - but where do they start?

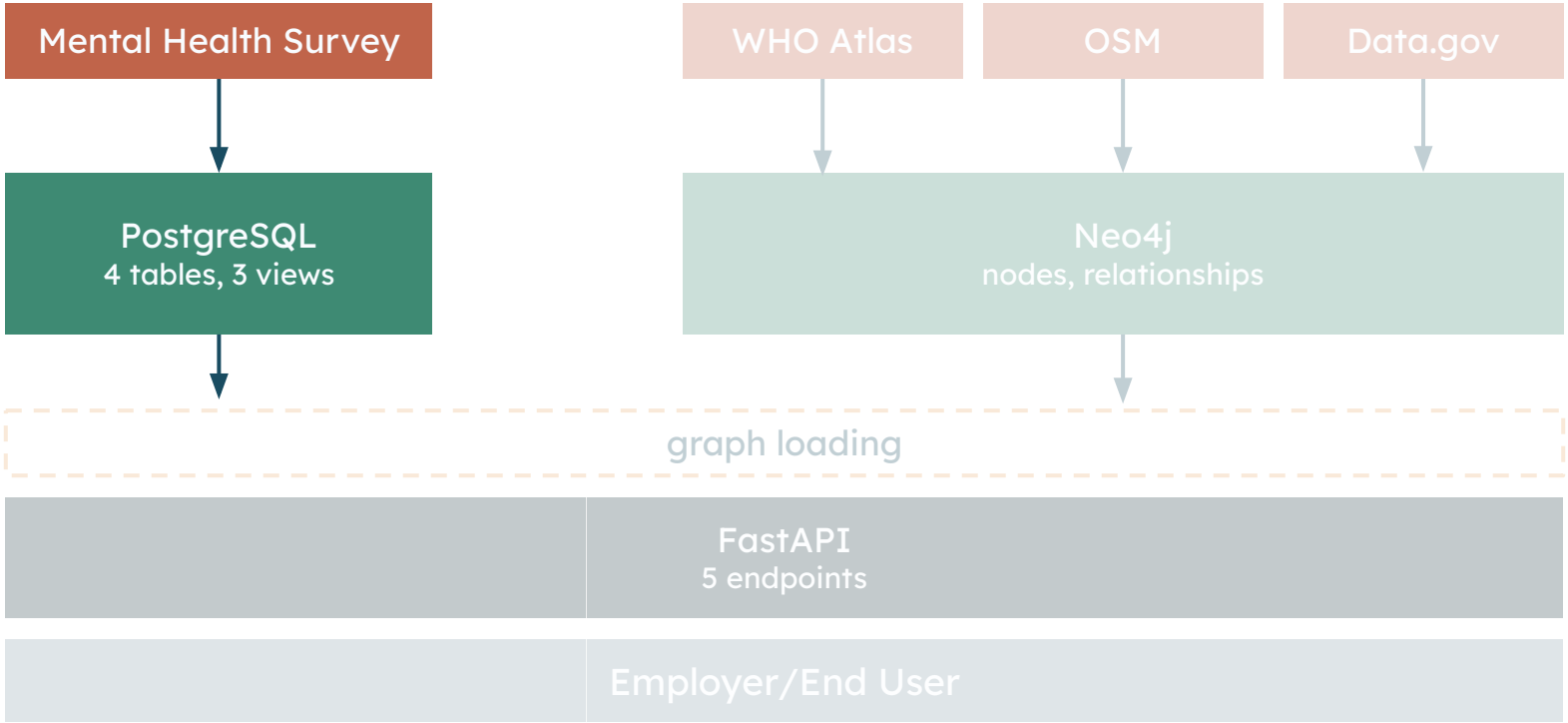
BridgeWell matches recommendations to real country-level data

[1] Based on survey responses

System Architecture



System Architecture: PostgreSQL



The Data

Mental Health Survey

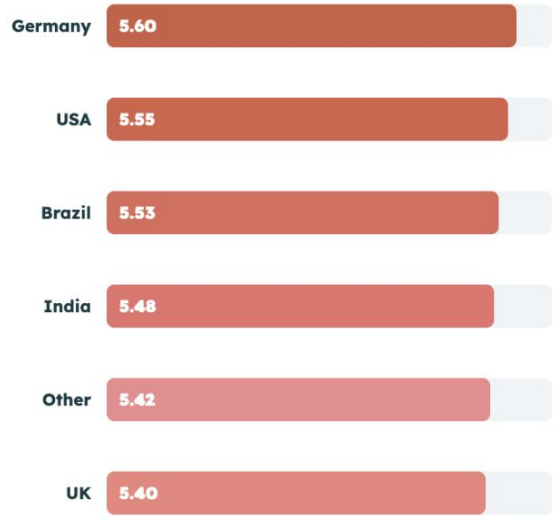
Powered by yFiles

Exploring the Data

global mental health crisis and treatment disparities

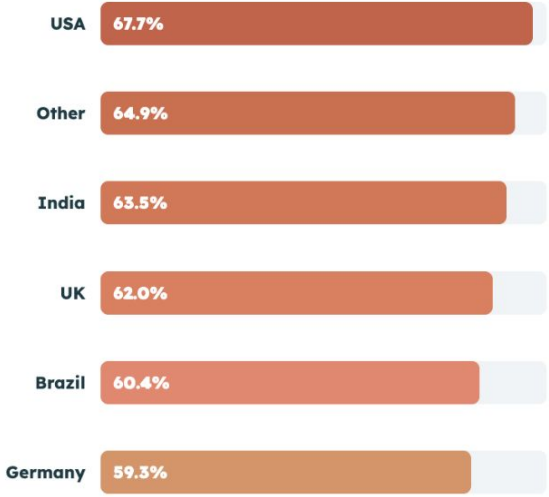
AVG STRESS SCORE POSTGRESQL

employer_gap_by_country → avg_stress



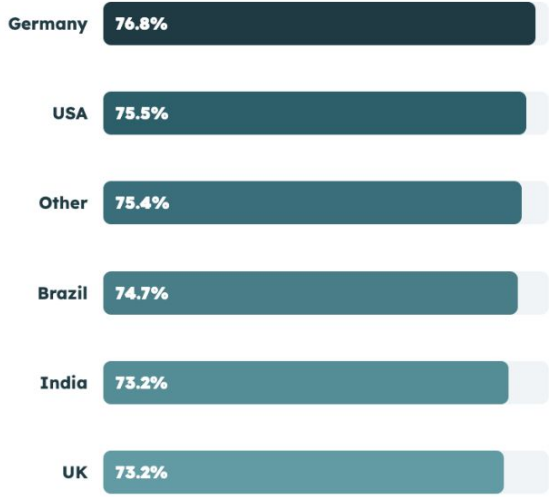
TREATMENT GAP POSTGRESQL

treatment_gap_by_country → treatment_gap_pct

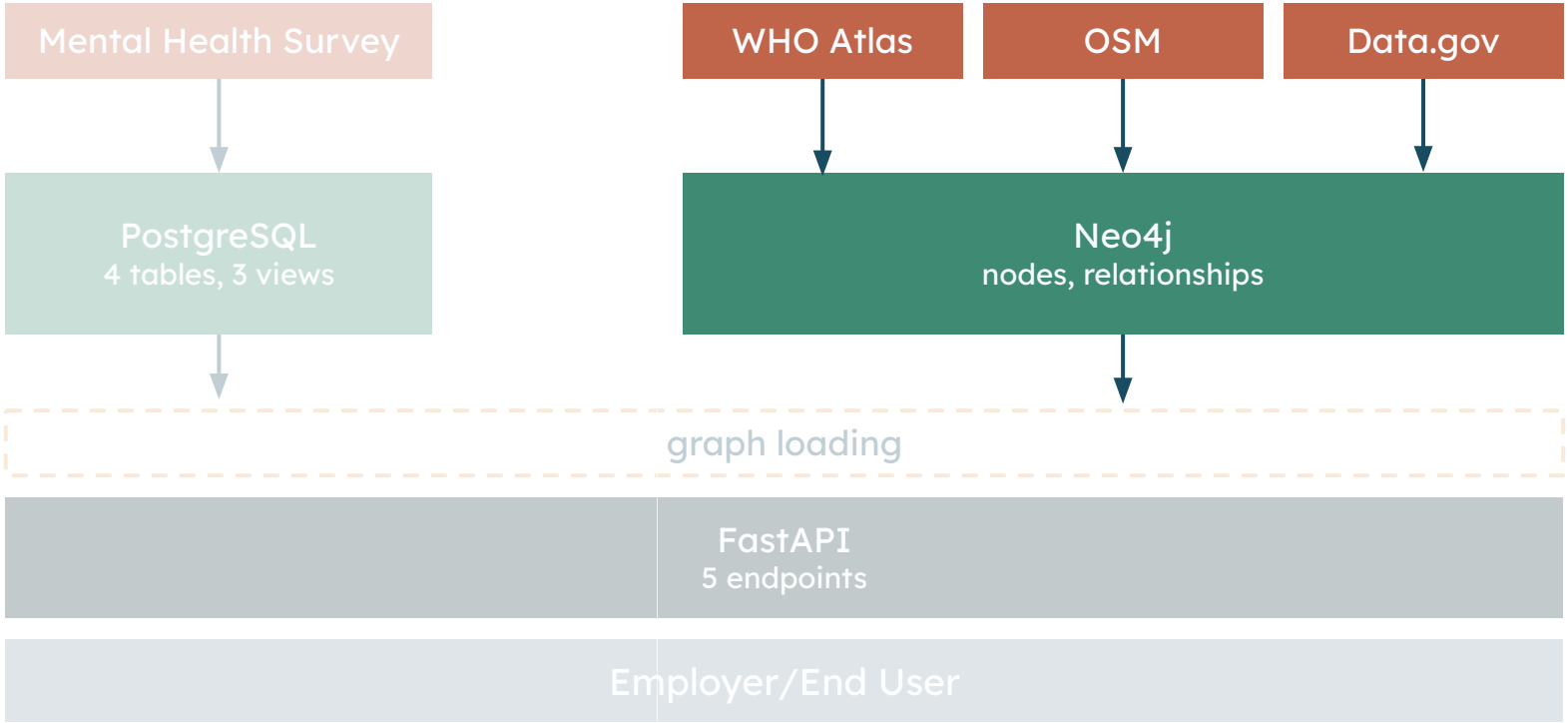


SUPPORT GAP POSTGRESQL

employer_gap_by_country → pct_lacking_support



System Architecture: Neo4j



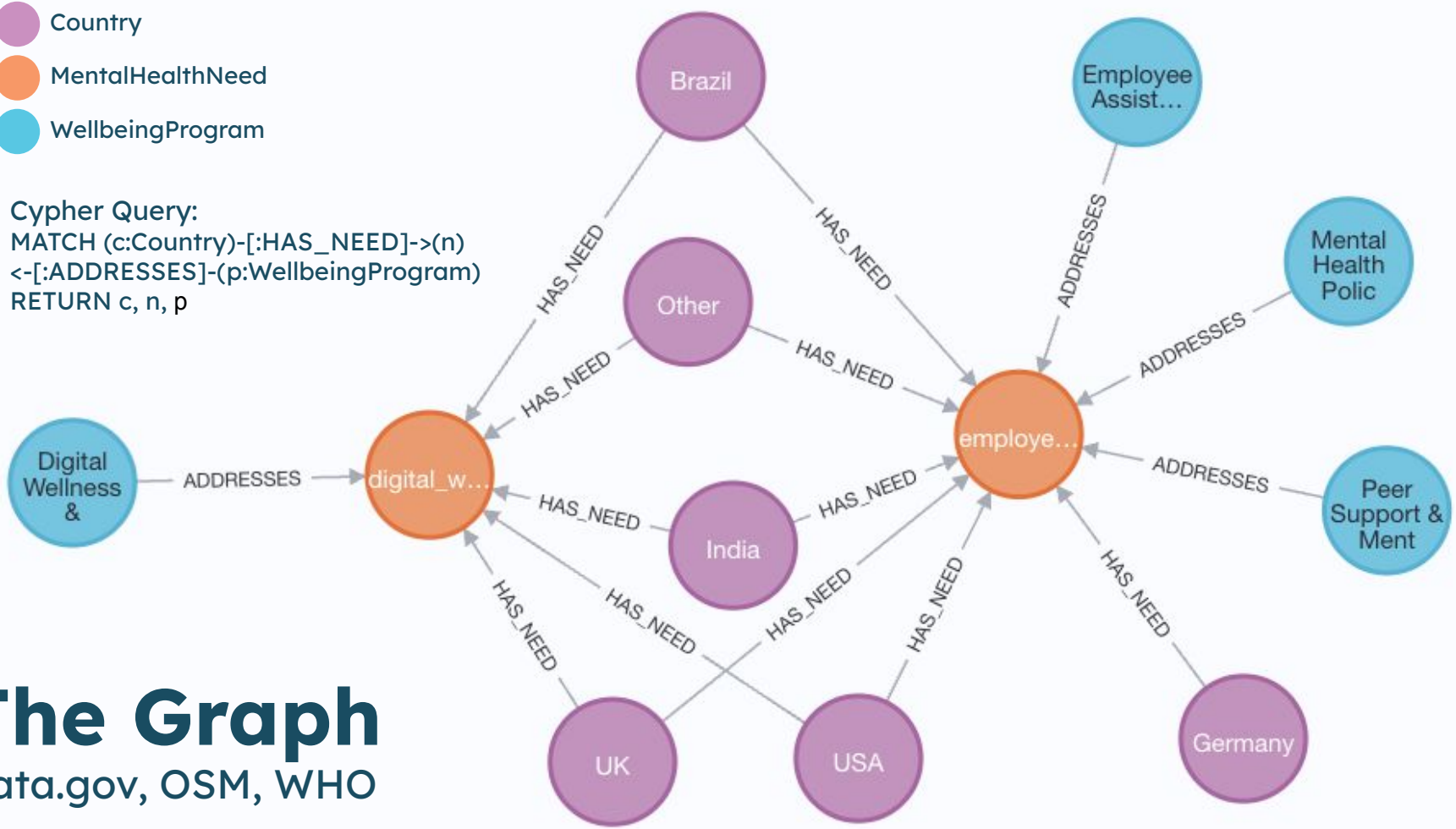
Country

MentalHealthNeed

WellbeingProgram

Cypher Query:

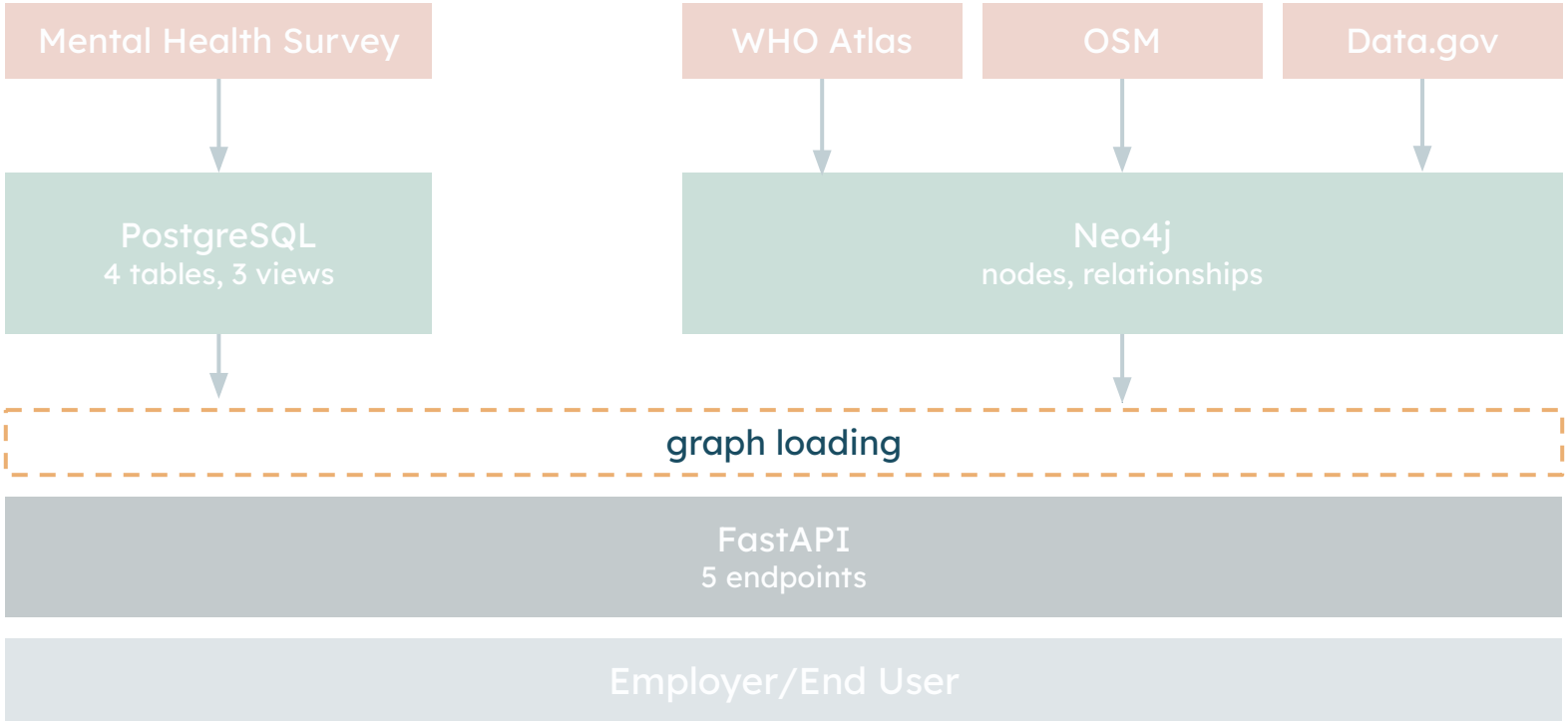
```
MATCH (c:Country)-[:HAS_NEED]->(n)
<-[:ADDRESSES]-(p:WellbeingProgram)
RETURN c, n, p
```



The Graph

Data.gov, OSM, WHO

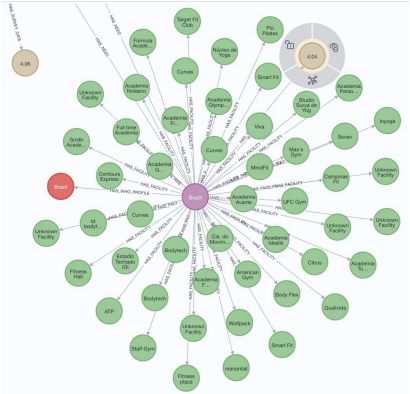
System Architecture: Graph Loading Pipeline



Graph Loading Pipeline

the database connection

- PostgreSQL aggregates become SurveySnapshot node property
- 3 APIs
  - MH access
  - Clinics & centers
  - Federal wellness plans
- Convert raw numbers to graph relationships



Node properties

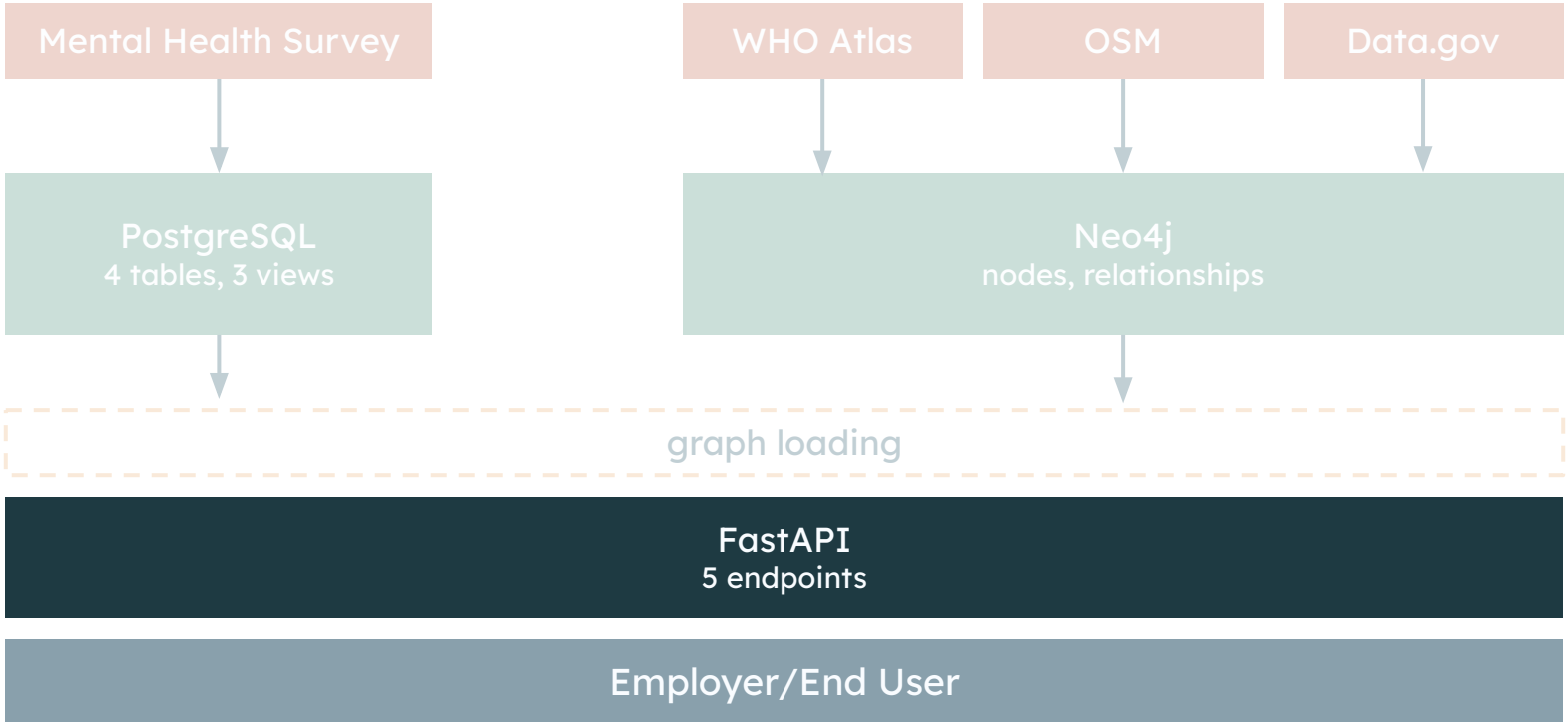
SurveySnapshot

<elementId>	4:d7a458e9-31fb-40b9-9efb-f1b4c757f0d2:5
<id>	5
avg_job_satisfaction	5.43
avg_screen_time	7.05
avg_sleep_hours	6.78
avg_stress	5.53
avg_symptom_score	4.04
avg_work_life_balance	5.58
country	Brazil
pct_lacking	74.7

Threshold Logic

avg_stress ≥ 6.0	→	stress_management need node
avg_symptom_score ≥ 5.0	→	anxiety_depression need node
avg_sleep_hours < 6.5	→	sleep_health need node
avg_screen_time ≥ 7.0	→	digital_wellness need node
pct_lacking_support ≥ 60%	→	employer_policy need node

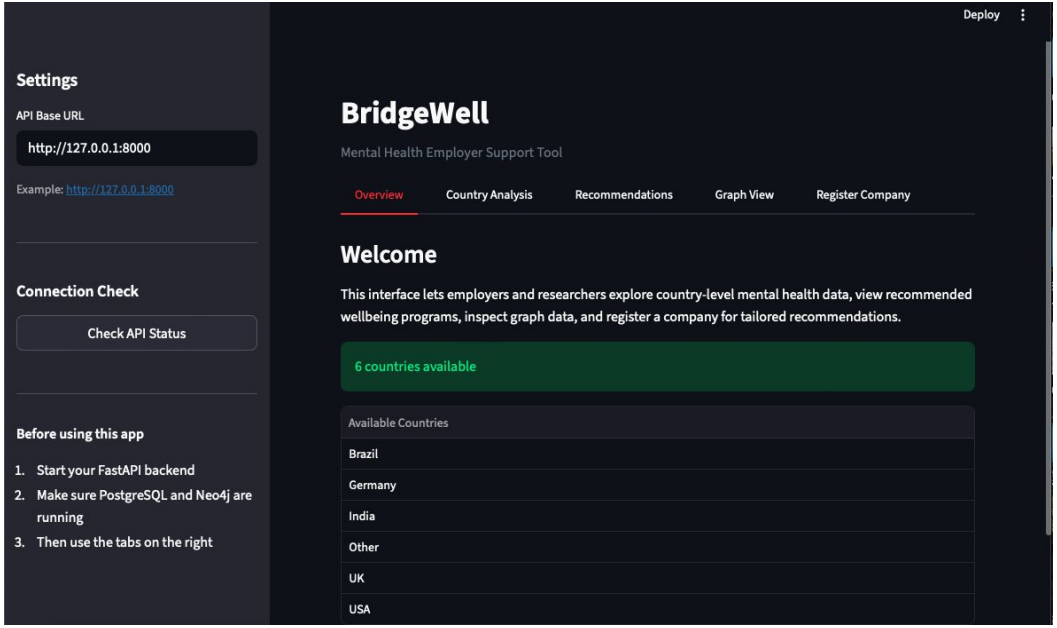
System Architecture: Graph Loading Pipeline



The API & UI: Company Input to Recommendations

five endpoints bridge PostgreSQL and Neo4j into single response

GET	/	Health
GET	/countries	List Countries
GET	/analysis/{country}	Country Analysis
GET	/recommendations/{country}	Get Recommendations
GET	/graph/{country}	Get Country Graph
POST	/company	Register Company



Country → Need → Program Recommendation Flow

Tailoring wellbeing programs based on identified mental health needs

Case: Germany

A mid sized tech company in Munich

TechCorp Munich

hybrid | medium | financial services

① API REQUEST POST /COMPANY

```
api/main.py → @app.post("/company")

# Request body
{
  "name":    "TechCorp Munich",
  "country": "Germany",
  "industry":"Technology",
  "size":    "medium",
  "remote_policy": "hybrid"
}

# Handler logic
async def register_company(body):
  # 1. Insert into PostgreSQL
  company_id = pg_query(
    "INSERT INTO companies ..."
  )
  # 2. Query SQL view
  context = pg_query(
    employer_gap_by_country,
    country=body.country
  )
  # 3. Traverse graph
  recs = neo4j_query(
    cypher, country=body.country,
    remote_only=False
  )
```

② SQL VIEW QUERY POSTGRESQL

```
sql/schema.sql → employer_gap_by_country

CREATE VIEW employer_gap_by_country AS
SELECT
  r.country,
  AVG(l.stress_level)
    AS avg_stress,
  COUNT(*) FILTER (
    WHERE NOT
      w.company_mental_health_support
  ) * 100.0 / COUNT(*)
    AS pct_lacking_support,
  AVG(m.composite_symptom_score)
    AS avg_symptom_score
FROM respondents r
JOIN work_profiles w    USING (respondent_id)
JOIN lifestyle_factors l USING (respondent_id)
JOIN mental_health_status m USING (respondent_id)
GROUP BY r.country;
```

RESULT — GERMANY (968 RESPONDENTS)

avg_stress	5.60 / 10 ↑ highest
pct_lacking_support	76.8% ↑ highest
avg_symptom_score	4.03 / 10
treatment_gap	59.3% untreated

③ GRAPH TRAVERSAL NEO4J CYPHER

```
api/main.py → neo4j_query()

MATCH
  (c:Country {name: "Germany"})
  -[:HAS_NEED]->
  (n:MentalHealthNeed)
  <-[:ADDRESSES]-
  (p:WellbeingProgram)
WHERE p.remote_ok = true
RETURN
  p.program_id,
  p.name,
  p.program_type,
  n.need_type
    AS addressed_need
ORDER BY
  p.coverage_score DESC
```

PEER_001	Peer Support & MH Champions	peer_support
POLICY_001	MH Policy & Manager Training	employer_policy
EAP_001	Employee Assistance Program	EAP

REQUEST FLOW POST /company → INSERT companies (PG) → pg_query(employer_gap_by_country) → neo4j_query(MATCH c:Country → n:Need ← p:Program) → 3 recommendations returned

Case: Brazil

A remote start-up in São Paulo

São Paulo Startup

remote only | small | technology

① API REQUEST POST /COMPANY

```
api/main.py → @app.post("/company")

# Request body
{
  "name":    "StartupSP",
  "country": "Brazil",
  "industry":"Technology",
  "size":    "small",
  "remote_policy": "yes"  ← key diff

}

# Same handler, different outcome:
# remote_policy:"yes" passes
# remote_only=True to neo4j_query
# → filters to remote_ok programs only
# → digital_wellness need active
# → DIGWEL_001 added to results

  recs = neo4j_query(
    cypher, country="Brazil",
    remote_only=True  ← activated

  )
```

② SQL VIEW RESULT POSTGRESQL

```
sql/schema.sql → treatment_gap_by_country

CREATE VIEW treatment_gap_by_country AS
SELECT
  r.country,
  COUNT(*) FILTER (
    WHERE m.has_mental_health_diagnosis
  ) AS diagnosed,
  COUNT(*) FILTER (
    WHERE m.has_mental_health_diagnosis
    AND NOT m.currently_in_treatment
  ) AS untreated,
  untreated * 100.0
    / NULLIF(diagnosed, 0)
    AS treatment_gap_pct
FROM respondents r
JOIN mental_health_status m
```

RESULT — BRAZIL (1,077 RESPONDENTS)

avg_stress	5.53 / 10
pct_lacking_support	74.7%
avg_symptom_score	4.04 / 10
treatment_gap	60.4% untreated

WHO: 4 psychiatrists/100k vs Germany's 14. MH budget 2.3% vs 11.3%. Same survey patterns, far weaker national infrastructure.

③ GRAPH TRAVERSAL NEO4J CYPHER

```
api/main.py → neo4j_query(remote_only=True)

MATCH
  (c:Country {name: "Brazil"})
  -[:HAS_NEED]->
  (n:MentalHealthNeed)
  <-[:ADDRESSES]-
  (p:WellbeingProgram)
WHERE p.remote_ok = true  ← filtered

RETURN
  p.program_id,
  p.name,
  p.program_type,
  n.need_type
    AS addressed_need
```

PEER_001	Peer Support & MH Champions	peer_support
POLICY_001	MH Policy & Manager Training	employer_policy
EAP_001	Employee Assistance Program	EAP
DIGWEL_001	Digital Wellness & Screen Time	digital_wellness +1 NEW

KEY DIFFERENCE remote_policy:"yes" → remote_only=True passed to Cypher WHERE clause → digital_wellness need node active in Brazil graph → 4 recommendations (not 3)

BridgeWell

EMPLOYEE SUPPORT SOLUTIONS

What should your company do next?